

Lithosols

Lithosols (also called “Leptosols” in some classification systems, though with slight differences) are very shallow soils overlying continuous hard rock or very coarse, stony parent material.

The term “Lithosol” reflects this constraint: “litho-” for rock, “-sol” for soil. They are defined by having very little fine earth above the rock — often continuous hard rock within about **10–25 cm** of the surface. Because of their thin profile, they have minimal horizon development; B-horizons are often absent or very weak. The soil is often stony, skeletal, or with large fragments mixed in the shallow topsoil. They occur in many climatic regions, especially where erosion is active or parent rock is resistant and weathering is slow.

Drainage:

- **Wet season:** Drainage tends to be **good to very good** superficially because there is little soil above the rock; water either runs off or quickly infiltrates where there are cracks in the rock or soil matrix. But because of minimal storage, the soil cannot buffer much water, so surface runoff can be high.
- **Dry season:** Very little moisture retention; soils dry quickly once rains stop. Vegetation often experiences drought stress.
- **Landscape:** Usually found on steep slopes, escarpments, ridges, and mountainous terrains. Also on shallow rocky plateaus.

Depth:

- Very shallow, usually **<10–25 cm** of fine earth before hitting bedrock or continuous hard rock. Some classifications use ≤ 10 cm cutoff.
- Rooting depth is severely restricted by shallow soil and hard substrate; roots often shallow, spread laterally rather than vertically.

Workability:

- **When wet:** The surface can become muddy or slick in patches (especially if there is some fine earth), but because the profile is shallow, tools can hit rock or bedrock; very difficult to cultivate heavy equipment.
- **When dry:** Soil is often thin, fragile, prone to crusting; mechanical tillage almost impossible; small hand tools more suited.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low, because of low fine earth content and often coarse texture; limited clay and organic matter.
- **Base saturation:** Depending on parent rock — if parent is rich in bases (e.g. basic igneous rock), base saturation might be moderate; often though poor due to low water and nutrient retention.
- **Organic matter:** Usually low; surface organic matter may accumulate where vegetation exists, but decomposition and accumulation are limited by shallow profile and exposure.
- **Nutrients:** Nitrogen and phosphorus are often deficient. Micronutrients may be variable depending on parent material.

pH:

- Can vary widely depending on parent rock. If rock is basic (e.g. limestone, basalt), pH may be near neutral or slightly alkaline. If acidic rock (granite, quartzite), pH may be acidic. Overall, no consistent pH — depends strongly on geology.

Management:

- Focus on **erosion control:** because soils are shallow and stony, erosion quickly removes the little topsoil that exists. Use terraces, check dams, retaining walls, or vegetation strips.
- Use shallow-rooted or drought-tolerant crops that do not require deep soils.
- Add organic matter where possible (mulch, compost) to build up the thin soil layer, improve moisture retention, and increase fertility in the surface horizon.
- Avoid deep tillage or heavy machinery which will disrupt or expose bedrock and cause degradation.
- Fertilizer use should be cautious; small, frequent applications may be better; focus on nutrients that are most limiting (often N, P). Soil testing helps.
- Plant grasses, shrubs or trees with shallow root systems to stabilize soil, prevent erosion, and gradually build soil depth.